charge of the lattice (see App. D),

$$
\begin{aligned}
H_{1h} &= \frac{h^2}{2}\sum_{a=1}^{8}\left(\widetilde{Q}^{(a)}\widetilde{Q}^{(a)}\otimes\widetilde{I}+\widetilde{I}\otimes\widetilde{\widetilde{Q}}^{(a)}\widetilde{\widetilde{Q}}^{(a)}+2\widetilde{Q}^{(a)}\otimes\widetilde{\widetilde{Q}}^{(a)}\right)\\
&= \frac{h^2}{2}\Bigg(16w_1\otimes w_1-\frac{8}{3}w_1\otimes(w_3+w_5+w_7)-\frac{8}{3}(w_3+w_5+w_7)\otimes w_1\\
&\quad-(w_3-w_5)\otimes(w_3-w_5)-\frac{1}{3}(w_4+w_6-2w_8)\otimes(w_4+w_6-2w_8)\\
&\quad+\frac{1}{2}\sum_{\substack{r\in\{(23),(24),(34),\\(56),(57),(67)\}}}(\mathcal{Y}_r\otimes\mathcal{Y}_r-\mathcal{X}_r\otimes\mathcal{X}_r)\\
&\quad+\frac{1}{2}\sum_{\substack{(r,s)\in\{(23)(56),\\(24)(57),(34)(67)\}}}(\mathcal{X}_r\otimes\mathcal{X}_s+\mathcal{Y}_r\otimes\mathcal{Y}_s+\mathcal{X}_s\otimes\mathcal{X}_r+\mathcal{Y}_s\otimes\mathcal{Y}_r)\Bigg)\ .
\end{aligned}
\tag{22}
$$

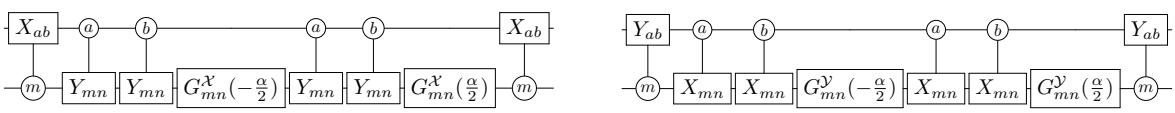

FIG. 4. Quantum circuits acting on qu8its that implement (left panel) $G^{\mathcal{XX}}_{abmn}(\alpha)=e^{-i\alpha\mathcal{X}_{ab}\otimes\mathcal{X}_{mn}}$, and (right panel) $G^{\mathcal{YY}}_{abmn}(\alpha)=e^{-i\alpha\mathcal{Y}_{ab}\otimes\mathcal{Y}_{mn}}$. These are required to implement Trotterized time evolution, along with single-qu8it gates, corresponding to the Hamiltonian terms in Eqs. (19), (20), (21) and (22). This circuit structure is reproduced from our previous work [66].

The Trotterized time evolution arising from this Hamiltonian, written in terms of generators of Givens transformation, can be implemented on qu8its using known quantum circuits, requiring 96 two-qu8it Givens rotations, and including the H_{1h} contribution would require an additional 26 rotations.[8] Figure 4 shows the quantum circuits for implementing the unitary operations of the form $G^{\mathcal{XX}}_{abmn}(\alpha)=e^{-i\alpha\mathcal{X}_{ab}\otimes\mathcal{X}_{mn}}$ and $G^{\mathcal{YY}}_{abmn}(\alpha)=e^{-i\alpha\mathcal{Y}_{ab}\otimes\mathcal{Y}_{mn}}$, which are the only required entangling structures, as a function of controlled gates. The associated gate-count for implementing one Trotter step of time evolution for $N_f=1$ and $L=1$ is, including the h-term, 732 controlled-X and -Y gates and 249 single qu8it rotations, which is much larger compared to the direct application of $G^{\mathcal{XX}}_{abmn}(\alpha)$ and $G^{\mathcal{YY}}_{abmn}(\alpha)$ via the generalized Mølmer-Sørensen (MS) gates [67, 94].

2. *Restructuring*

A recent paper [82], where a different route to implementing similar types of gates is taken, indicates significant reductions in two-qu8it rotations can be gained by grouping operators. This is motivated by advances in quantum hardware [67, 68] so that multiple of the transitions associated with, for instance, the addition of a red-quark at a given site can be implemented simultaneously, as opposed to sequentially. This suggests that all of the operations associated with the red-quark-creation operator can be grouped together. Similarly for the green and blue operations, and for their corresponding annihilation. The kinetic-energy term in Eq. (19) can be greatly simplified and written as

$$
H_{1kin}=\frac{1}{4}(A_0^{(r)}\otimes A_1^{(r)}-B_0^{(r)}\otimes B_1^{(r)})+\frac{1}{4}(A_0^{(g)}\otimes A_1^{(g)}-B_0^{(g)}\otimes B_1^{(g)})+\frac{1}{4}(A_0^{(b)}\otimes A_1^{(b)}-B_0^{(b)}\otimes B_1^{(b)})\ , \tag{23}
$$

[8] We had hoped that the qu8it mapping would eliminate or mitigate the violation of SU(3) color charge occurring for $L>1$ lattices due to Trotterization (that we identified in Ref. [37]). However, this is not the case, and Trotterization of the time-evolution arising from the qu8its Hamiltonian also violates color-charge conservation for $L>1$.

where

$$
\begin{aligned}
A_0^{(r)} &= \mathcal{X}_{(12)} - \mathcal{X}_{(37)} + \mathcal{X}_{(46)} + \mathcal{X}_{(58)} \ , \quad A_1^{(r)} = \mathcal{X}_{(12)} + \mathcal{X}_{(37)} - \mathcal{X}_{(46)} + \mathcal{X}_{(58)} \ , \\
A_0^{(g)} &= \mathcal{X}_{(13)} + \mathcal{X}_{(27)} - \mathcal{X}_{(45)} + \mathcal{X}_{(68)} \ , \quad A_1^{(g)} = \mathcal{X}_{(13)} - \mathcal{X}_{(27)} + \mathcal{X}_{(45)} + \mathcal{X}_{(68)} \ , \\
A_0^{(b)} &= \mathcal{X}_{(14)} - \mathcal{X}_{(26)} + \mathcal{X}_{(35)} + \mathcal{X}_{(78)} \ , \quad A_1^{(b)} = \mathcal{X}_{(14)} + \mathcal{X}_{(26)} - \mathcal{X}_{(35)} + \mathcal{X}_{(78)} \ ,
\end{aligned} \tag{24}
$$

and the $B_n^{(\alpha)}$ operators are analogous to the $A_n^{(\alpha)}$ operators with $\mathcal{X} \leftrightarrow \mathcal{Y}$. The 96 two-qu8it entangling terms in Eq. (19) are reduced to just 6 in Eq. (23). The mass term is unchanged, as is the contribution from the chromo-electric field proportional to g^2, neither of which involve two-qu8it entangling gates.

For the h-term, a similar simplification of terms exists by groupings into commuting sets,

$$
\begin{aligned}
H_{1h} = \frac{h^2}{2} \ \Bigg(& \frac{1}{2}(D^{(12)} \otimes D^{(12)} - C^{(12)} \otimes C^{(12)}) + \frac{1}{2}(D^{(45)} \otimes D^{(45)} - C^{(45)} \otimes C^{(45)}) \\
& + \frac{1}{2}(D^{(67)} \otimes D^{(67)} - C^{(67)} \otimes C^{(67)}) + 2\widetilde{Q}^{(3)} \otimes \widetilde{\widetilde{Q}}^{(3)} + 2\widetilde{Q}^{(8)} \otimes \widetilde{\widetilde{Q}}^{(8)} \\
& + 2\widetilde{I} \otimes \widetilde{I} - \mathrm{diag}\left(1, -\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, 1\right) \otimes \widetilde{I} - \widetilde{I} \otimes \mathrm{diag}\left(1, -\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, 1\right)\Bigg) \ .
\end{aligned} \tag{25}
$$

where the C and D terms are, respectively,

$$
\begin{aligned}
C^{(12)} &= \mathcal{X}_{(23)} - \mathcal{X}_{(56)} \ , \quad C^{(45)} = \mathcal{X}_{(24)} - \mathcal{X}_{(57)} \ , \quad C^{(67)} = \mathcal{X}_{(34)} - \mathcal{X}_{(67)} \ , \\
D^{(12)} &= \mathcal{Y}_{(23)} + \mathcal{Y}_{(56)} \ , \quad D^{(45)} = \mathcal{Y}_{(24)} + \mathcal{Y}_{(57)} \ , \quad D^{(67)} = \mathcal{Y}_{(34)} + \mathcal{Y}_{(67)} \ .
\end{aligned} \tag{26}
$$

The 26 two-qu8it operations in H_{1h} in Eq. (22) is reduced to 8 in Eq. (25) by this restructuring.

B. QCD with N_f Flavors on L Spatial Site and Quantum Resource Requirements

Considering the general situation of an arbitrary number of spatial lattice sites and an arbitrary number of flavors of quarks, the Hamiltonian becomes (including the h-term),

$$
\begin{aligned}
\hat{H} &= \hat{H}_{kin} + \hat{H}_m + \hat{H}_{el} + \hat{H}_h \ , \\
&= \frac{1}{2} \sum_{n=0}^{2L-2} \sum_{f=u,d,s,\ldots} \sum_{\alpha=r,g,b} \left(\hat{c}^\dagger_{\alpha,f,n} \hat{c}^\dagger_{\alpha,f,n+1} - \hat{c}_{\alpha,f,n} \hat{c}_{\alpha,f,n+1} \right) + 3 \sum_{n=0}^{2L-1} \sum_{f=u,d,s,\ldots} m_f \ \hat{\mathcal{B}}_{f,n} \\
&\quad + \frac{g^2}{2} \sum_{n=0}^{2L-2} \sum_{a=1}^{8} \left(\sum_{m \le n} \sum_{f=u,d,s,\ldots} \hat{Q}^{(a)}_{f,m} \right)^2 + \frac{h^2}{2} \sum_{n=0}^{2L-1} \sum_{a=1}^{8} \left(\sum_{m \le n} \sum_{f=u,d,s,\ldots} \hat{Q}^{(a)}_{f,m} \right)^2 \ .
\end{aligned} \tag{27}
$$

The mapping to qu8its is a generalization of that discussed above, and displayed in Fig. 1. There are $2N_f L$ qu8its, half support the quark sites and half support the anti-quark sites. The kinetic term connects adjacent qu8its and anti-qu8its of the same flavor across the lattice, the mass term provides contributions from individual qu8its and anti-qu8its, and the chromo-electric term(s) connects all qu8its and anti-qu8its. The kinetic term in the Hamiltonian becomes,

$$
H_{kin} = \sum_{n=0}^{2L-2} \sum_{f=u,d,s,\ldots} \sum_{\alpha=r,g,b} \frac{1}{4} (A^{(\alpha)}_{0,f,n} \otimes A^{(\alpha)}_{1,f,n+1} - B^{(\alpha)}_{0,f,n} \otimes B^{(\alpha)}_{1,f,n+1}) \ , \tag{28}
$$

where $A^{(\alpha)}_{0,f,n}$ denotes $A^{(\alpha)}_0$ from Eq. (24), acting on the flavor f qu8it at site n. The mass terms given in Eq. (27), with a straightforward generalization of the above, become

$$
H_m = 3 \sum_{n=0}^{2L-1} \sum_{f=u,d,s,\ldots} m_f \ \widetilde{\mathcal{B}}_{f,n} \ . \tag{29}
$$

Qudits	Number of qudits	U_{kin} ent. gates	U_{el} ent. gates
Qubit ($d=2$)	$6N_fL$	$6N_f(8L-3)-4$	$N_f(2L-1)[23N_f(2L-1)-17]$
Qu8it ($d=8$)	$2N_fL$	$6N_f(2L-1)$	$4N_f(2L-1)[N_f(2L-1)-1]$
Reduction in resources ($L\to\infty$)	3	4	5.75

TABLE I. The number of qudits ($d=2$ and $d=8$) and entangling gates for applying a single Trotter step using the unitary operators corresponding to the kinetic (U_{kin}) and the $\mathcal{O}(g^2)$ chromo-electric (U_{el}) parts of the SU(3) Hamiltonian, comparing qubit and qu8it implementations.

The color charge-charge contribution is somewhat more involved due to the number of terms, but can be simplified using symmetries of the sums, as shown in App. D. The summation can be reorganized,

$$\begin{aligned}\sum_{n=0}\sum_{a=1}^{8}\left(\sum_{m\leq n}\sum_{f=u,d,s,\ldots}\hat{Q}^{(a)}_{f,m}\right)^2 &= \sum_{n=0}\sum_{m,m'\leq n}\sum_{f,f'}\sum_{a=1}^{8}\hat{Q}^{(a)}_{f,m}\hat{Q}^{(a)}_{f',m'}\\ &= \sum_{n=0}\sum_{\substack{m,m'\in\text{even}\\ m,m'\in\text{odd}}}^{n}\sum_{f,f'}\left(\sum_a \widetilde{Q}^{(a)}_{f,m}\,\widetilde{Q}^{(a)}_{f',m'}\right) + 2\sum_{n=0}\sum_{\substack{m\in\text{even}\\ m'\in\text{odd}}}^{n}\sum_{f,f'}\left(\sum_a \widetilde{Q}^{(a)}_{f,m}\,\widetilde{\widetilde{Q}}^{(a)}_{f',m'}\right),\end{aligned} \tag{30}$$

where some of the terms act on the same site and flavor. Equation (30) can be expanded in terms of the D and C operators from Eq. (26).

Now that the full Hamiltonian has been decomposed into one-qu8it and two-qu8it operations, an estimate of quantum resources can be performed for a single Trotter step of time evolution for the qu8it mappings, and compared to previously established results for qubit mappings [37]. Since the Trotter decomposition used in this work is the same as the one used for qubits [37], the error accumulated after performing a certain amount of Trotter steps will be the same (see Fig. 14 in Ref. [37]). The advantage of using qu8its will be in the reduction of gates required to apply a single Trotter step. Table I displays the resource estimates for entangling gates for qu8its and qubits. The number of entangling operations required for the qu8it mapping is significantly less than for the qubit mapping (by factors $\gtrsim 5$).

V. SUMMARY AND OUTLOOK

Motivated by continuing advances in the development of qudits for quantum computing, we have explored mapping 1+1D QCD to $d=8$ qudits. We have presented the general framework for performing quantum simulations of QCD with arbitrary number of flavors and lattice sites, and provided a detailed discussion of the theory with $N_f=1$ and $L=1$. The main reason for considering performing quantum simulations using qu8its is because the number of two-qu8it entangling operations required to evolve a given state forward in time is significantly less (more than a factor of 5 reduction) than the corresponding number for mappings to qubits. This is an important consideration for two main reasons. One is that the time to perform a two-qudit entangling operation on a quantum device is much longer than for a single qudit operation, and the second is the relative fidelity of the two types of operations. The naive mapping with sequentially-Trotterized entangling operations does not provide obvious gains, but the recently developed capabilities to simultaneously induce multiple transitions within qudits, enabling multiple entangling operations to be performed in parallel, is the source of the large gain. Thus, qudit devices of comparable fidelity gate operations and coherence times to an analogous device with a qubit register, are expected to be able to perform significantly superior quantum simulations of 1+1D QCD.

The results presented in this work readily generalize to an arbitrary number of colors. For the $N_c=2$ case, relevant for $SU(2)$, ququarts ($d=4$) are needed to embed the vacuum in 1 state, single quarks in 2 states, and singlet two-quark in 1 state. The number of entangling gates for each term of the kinetic piece of the Hamiltonian is reduced to 4, and for each $\widetilde{Q}^{(a)}\otimes\widetilde{Q}^{(a)}$ term, 3 entangling gates are required. For $N_c=4$, analogous gains can be achieved using qudits with $d=16$, qu16its. The mapping is such that the vacuum occupies 1 state, single quarks occupy 4, two quarks occupy 6, three quarks occupy 4, and four quarks occupy 1. It requires 8 entangling gates for the kinetic piece, and 15 for each $\widetilde{Q}^{(a)}\otimes\widetilde{Q}^{(a)}$ term. Quarks transforming in higher-dimension gauge groups can be mapped in similar ways, with $2N_c$ terms needed for the kinetic piece, and N_c^2-1 for $\widetilde{Q}^{(a)}\otimes\widetilde{Q}^{(a)}$. While the reduction in